Assignment 7

This exercise has two parts:

- Part A consists of mandatory exercises; you need to complete these and hand in your answers to be able to partake in the exam. This part also provides you with an impression of the minimum knowledge level necessary to pass the exam.
- Part B consists of additional exercises. These repeat a concept that is covered in Part A or help to broaden or deepen your understanding of some of the topics.

Assignment by

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Part A - Mandatory Exercises

Below you will find the mandatory exercises for Week 7 of the course Data Mining and Machine Learning.

Exercise 1 - AUC or Accuracy?

We explore the problem of class imbalance, and compare a classifier to randomly picking a label (which can be surprisingly effective in terms of accuracy!).

a. Based on the labels and probabilities below, calculate the AUC. Additionally, calculate the AUC for a simple classifier that randomly guesses (in other words, assigns probability 0.5 to each data point).

If you wish, you can use the `sklearn.metrics.roc_auc_score` function to calculate the AUC.

```
In [5]: import numpy as np
        from sklearn.metrics import roc_auc_score

        Labels = np.array([0,0,0,1,0,1,1,1,1,1,0,1,1,1,1,0,1,1,1,1,1,1,1,0,1,1,1,1,1,1,1,1,1,1,1,1,1,1,0,1,1,1,1,1])
        Classifier_1_probs = np.array([0.01,0.03,0.06,0.08,0.11,0.13,0.15,0.18,0.2,0.23,0.25,0.27,0.30,0.32,0.34,0.37,0.39,0.42,0.44,0.46,0.49,0.51,0.52,0.53,
        0.54,0.56,0.58,0.61,0.63,0.66,0.68,0.7,0.73,0.75,0.77,0.8,0.82,0.85,0.87,0.89,0.92,0.94,0.97,0.99])

        auc_score_Classifier_1 = roc_auc_score(Labels, Classifier_1_probs)
        auc_score_Random_guessing = 0.5

        print(auc_score_Classifier_1)
```

```
0.7743055555555556
```

b. Is Classifier 1 better or worse than the random guessing classifier in terms of AUC?

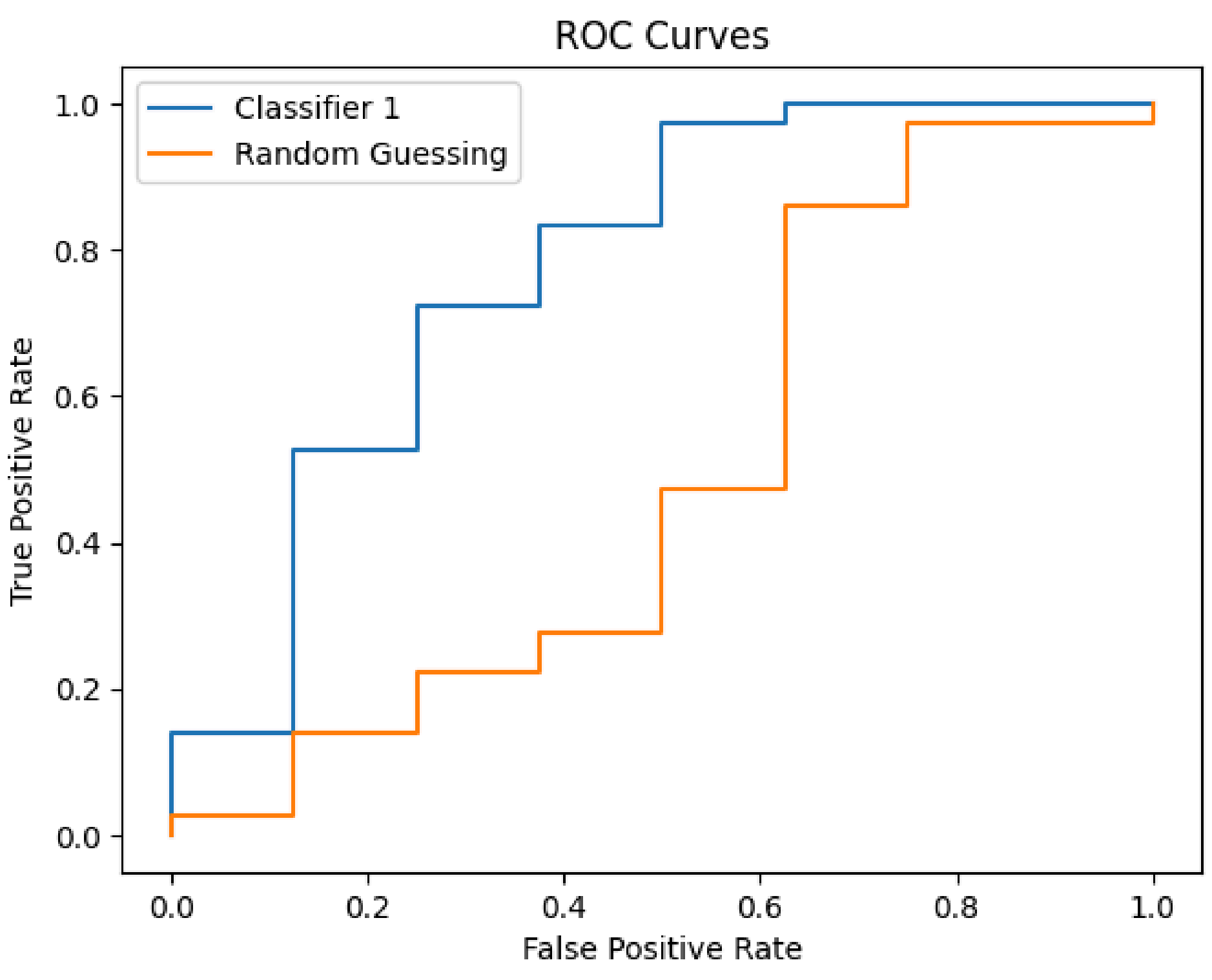
ANSWER: Classifier 1 is better than the random guessing classifier, since it has a higher AUC score.

c. Calculate the ROC curves for both of the classifiers by calculating the FP and TP rates and plot them in the same plot. Make the plot yourself as opposed to using a plotting library.

Hint: the function `sklearn.metrics.roc_curve` can be used for computing the FP and TP rates. You can then plot them using the `matplotlib.pyplot` plot function. Remember to include a legend.

```
In [8]: import matplotlib.pyplot as plt
        from sklearn.metrics import roc_curve

        fpr1, tpr1, _ = roc_curve(Labels, Classifier_1_probs)
        fpr_random, tpr_random, _ = roc_curve(Labels, np.random.rand(len(Labels)))
        plt.plot(fpr1, tpr1, label='Classifier 1')
        plt.plot(fpr_random, tpr_random, label='Random Guessing')
        plt.xlabel('False Positive Rate')
        plt.ylabel('True Positive Rate')
        plt.title('ROC Curves')
        plt.legend()
        plt.show()
```



d. We are now going to classify all points. Set the threshold of both classifiers to 0.5. To be more precise: assign the predicted label to Class 1 if the predicted probability is equal to or larger than 0.5 and to Class 0 otherwise. Report the accuracies for both classifiers and report which one is largest.

```
In [10]: Classifier_1_predicted_labels = None
         Random_guess_predicted_labels = None

         threshold = 0.5
         #YOUR CODE HERE
```

e. Can you explain why the random guess model appears to be doing better in terms of accuracy than Classifier 1, even though it had a lower AUC? Does this mean that Classifier 1 is useless? Can you think of a situation where Classifier 1 might be more beneficial than always picking the same label?

f. In a bit of a different scenario, you are advising a client about a classification model that differentiates between 2 classes in your dataset. The accuracy of the model in question is high (95%), however there is major class imbalance present in the dataset. You have tried to explain to your client that although the accuracy of the model seems high, it might not actually be that good.

You suggest to use the AUC to evaluate model performance. However, your client wants to receive the accuracy measure, and does not care one single bit about the AUC. What baseline do you compare the accuracy of your model to, in order to make sure that the model is at all useful?

Exercise 2 - From the book!

Make Exercise 13 from Section 3.11.

ANSWER:

(a):

Based on the accuracies shown in Table 3.7 I would choose model T_{100}, since it seems that for A has better accuracy.

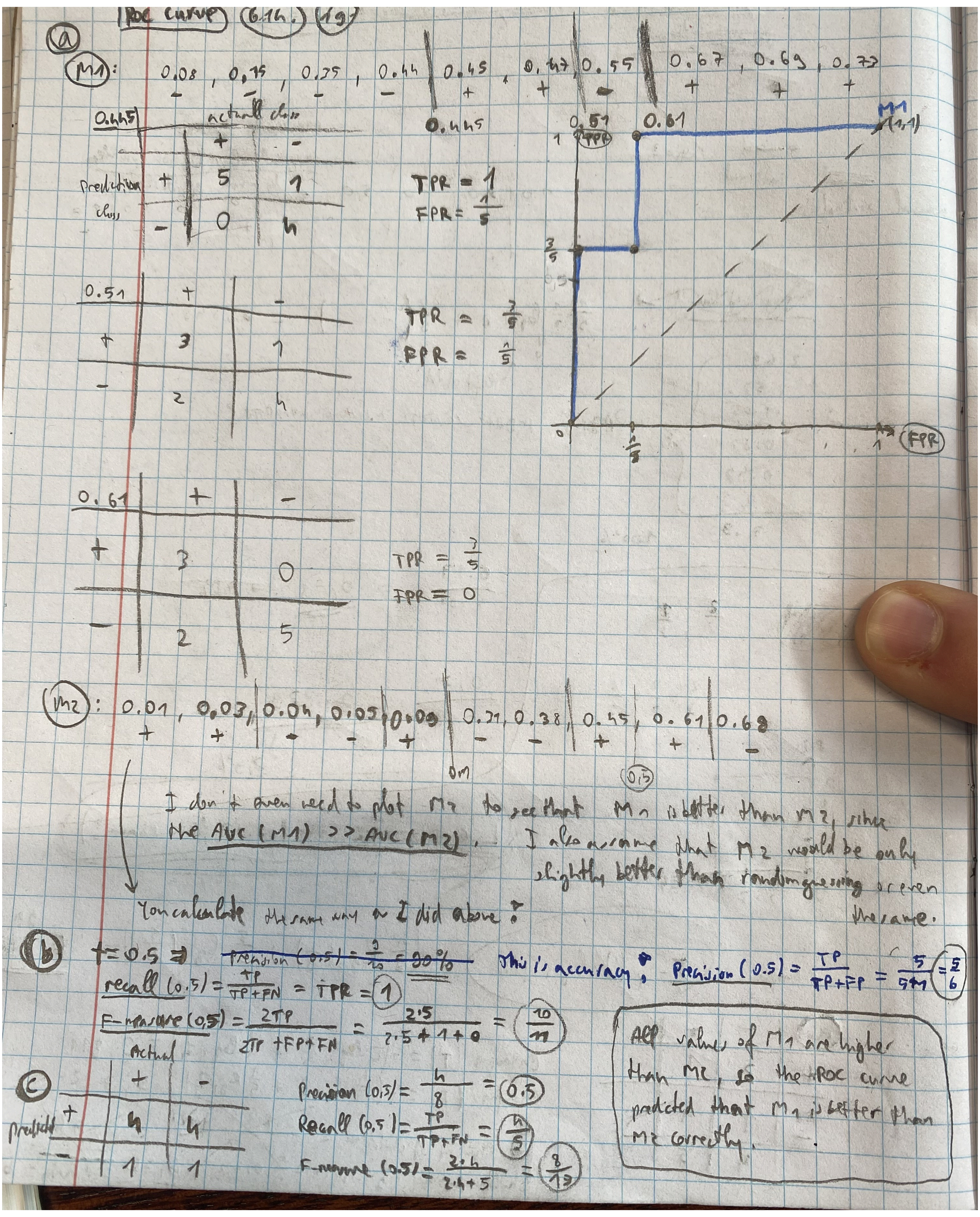
(b):

Since the classification accuracy of T_{10} and T_{100} don't differ much, I would choose T_{10}, since it's more consistent, so for unseen instances we can better predict what the accuracy would be, whereas for T_{100} if would appear that it is overfitted on A since the accuracy on B is much lower. This means, that it is most likely to drop further if another set C is introduces (new instances).

Exercise 3 - ...and another one

Make Exercise 19 from Section 6.14.

Note that you can of course make ROCs using any tool you like. However, it can be instructive to try make at least one ROC by hand.


M1
(1,1)
0.51
0.61
1
TPR
3/5
0
1/5
1
FPR

ROC curve (6.14.) 19

ⓐ

M1: 0,08 , 0,15 , 0,25 , 0,44 , 0,45 , 0,47 , 0,55 , 0,67 , 0,69 , 0,77
− − − − + + − + + +

0.445	actual class +	−
prediction class +	5	1
−	0	4

TPR = 1
FPR = $\frac{1}{5}$

0.51	+	−
+	3	1
−	2	4

TPR = $\frac{3}{5}$
FPR = $\frac{1}{5}$

0.61	+	−
+	3	0
−	2	5

TPR = $\frac{3}{5}$
FPR = 0

M2: 0.01 , 0.03 , 0.04 , 0.05 , 0.09 , 0.31 , 0.38 , 0.45 , 0.61 , 0.68
+ + − − + − − + + −

0.5

I don't even need to plot M2 to see that M1 is better than M2, since the AUC (M1) >> AUC (M2). I also assume that M2 would be only slightly better than random guessing or even the same.

You calculate the same way as I did above?

ⓑ $t = 0.5 \Rightarrow$ ~~Precision (0.5) = $\frac{9}{10}$ = 90%~~ This is accuracy! Precision (0.5) = $\frac{TP}{TP+FP} = \frac{5}{5+1} = \boxed{\frac{5}{6}}$

recall (0.5) = $\frac{TP}{TP+FN}$ = TPR = $\boxed{1}$

F-measure (0,5) = $\frac{2TP}{2TP+FP+FN} = \frac{2\cdot 5}{2\cdot 5+1+0} = \boxed{\frac{10}{11}}$

ⓒ

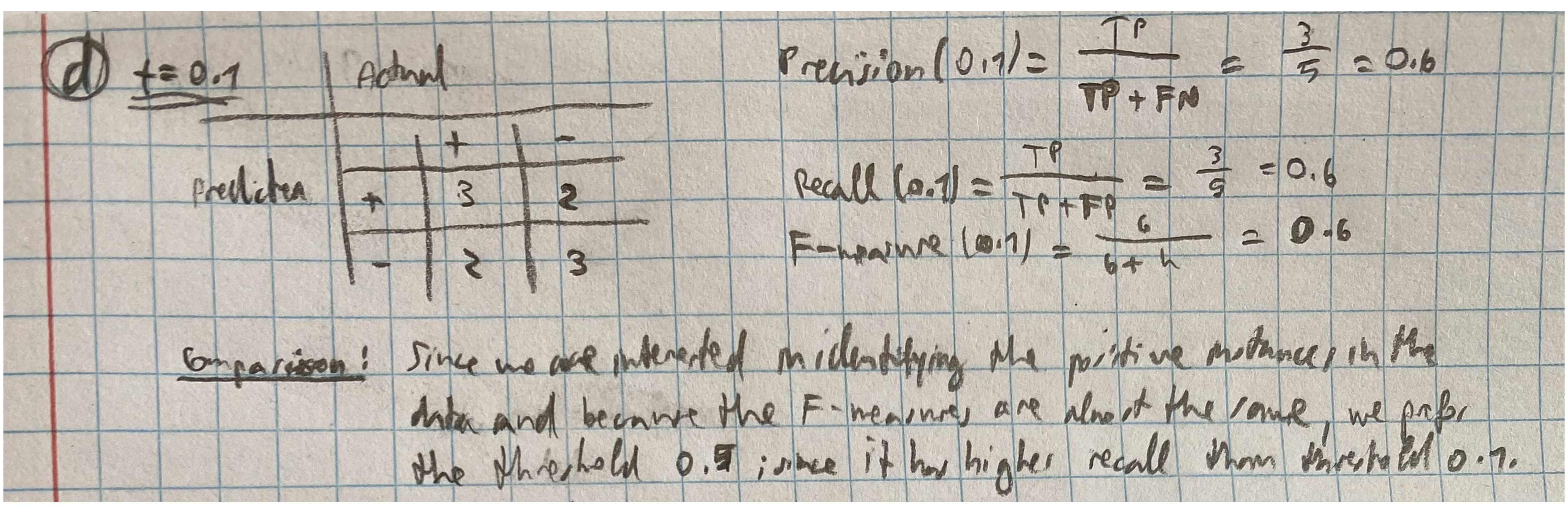
	Actual +	−
predicted +	4	4
−	1	1

Precision (0.5) = $\frac{4}{8} = \boxed{0.5}$

Recall (0.5) = $\frac{TP}{TP+FN} = \boxed{\frac{4}{5}}$

F-measure (0.5) = $\frac{2\cdot 4}{2\cdot 4+5} = \boxed{\frac{8}{13}}$

All values of M1 are higher than M2, so the ROC curve predicted that M1 is better than M2 correctly.



ⓓ $t = 0.1$

	Actual +	−
predicted +	3	2
−	2	3

Precision (0.1) = $\frac{TP}{TP+FN} = \frac{3}{5} = 0.6$

Recall (0.1) = $\frac{TP}{TP+FP} = \frac{3}{5} = 0.6$

F-measure (0.1) = $\frac{6}{6+4} = 0.6$

Comparison: Since we are interested in identifying the positive instances in the data and because the F-measures are almost the same, we prefer the threshold 0.5; since it has higher recall than threshold 0.1.

Part B

Exercise 4

Make Exercise 12 from TSKK, Section 3.11.